Slide Set 6: Diffusion Models & Flow Matching

Wasserstein Distance · DDPM · Score Matching · Flow Matching

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Outline

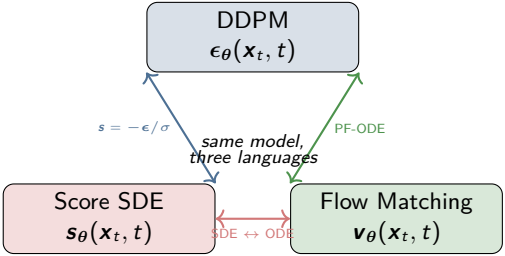
1. Distances Between Distributions
2. Denoising Diffusion Probabilistic Models
3. Score-Based Generative Models
4. Flow Matching
5. The Unifying Framework
6. Summary

Three Perspectives on Modern Generative Models

Central question: how to *measure* and *minimize* the gap between a model p_θ and data p_{data}?

Three complementary approaches, all converging on the same family:

- **Variational (DDPM)** — hierarchical VAE with a *fixed* noising encoder; loss = noise regression
- **Score-based** — estimate $\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$; sample via Langevin / reverse SDE
- **Flow-based** — learn a velocity field via Neural ODE; simulation-free training via CFM



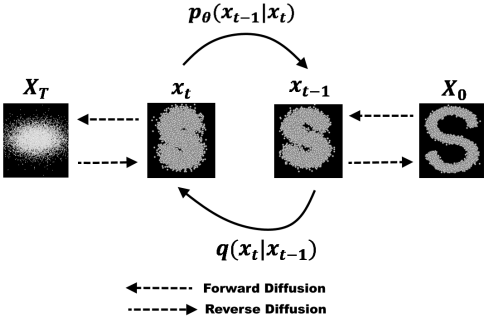
From VAE to DDPM

VAE recap: encoder $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ and decoder $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ are both *learned* jointly.

Hierarchical VAE: stack multiple latent layers $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_L$ for coarse-to-fine generation. Powerful, but training is unstable.

DDPM (Ho et al. 2020):

- **Fix the encoder** as a predefined Gaussian noising chain, no parameters to learn
- **Learn only the decoder** — a neural network that reverses the process step by step



DDPM: Forward Process

Forward Markov chain (fixed, $t = 1, \ldots, T$):

$$q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}) = \mathcal{N}(\boldsymbol{x}_t;\ \sqrt{\alpha_t}\,\boldsymbol{x}_{t-1},\ (1-\alpha_t)\mathbf{I})$$

with $\alpha_t = 1 - \beta_t$ and noise schedule $0 < \beta_1 < \cdots < \beta_T < 1$.

One-step update:

$$\boldsymbol{x}_t = \sqrt{\alpha_t}\,\boldsymbol{x}_{t-1} + \sqrt{1-\alpha_t}\,\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

At each step: scale down signal, inject Gaussian noise.

Closed-Form Marginal (Key Trick)

Define $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. By induction:

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) = \mathcal{N}\big(\boldsymbol{x}_t;\ \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big)$$

Reparameterization:

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

As $T \to \infty$ with appropriate schedule, $q(\boldsymbol{x}_T|\boldsymbol{x}_0) \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$: the clean signal is completely destroyed.

DDPM: Reverse Process

Goal: learn $p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)$ to reverse the noising chain.

Challenge: the true reverse $q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) \propto q(\boldsymbol{x}_t|\boldsymbol{x}_{t-1})\, q(\boldsymbol{x}_{t-1})$ requires integrating over p_{data} — intractable.

Conditioning Trick

Condition on $\boldsymbol{x}_0$: the posterior

$$q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0) = \mathcal{N}\big(\boldsymbol{x}_{t-1};\ \tilde{\boldsymbol{\mu}}_t,\ \tilde{\beta}_t \mathbf{I}\big)$$

is *Gaussian with closed form!*

During training $\boldsymbol{x}_0$ is always available, so this posterior can be evaluated exactly — giving us exact KL targets for each denoising step.

Posterior parameters:

$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\,\boldsymbol{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\,\boldsymbol{x}_t$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t$$

Parametric reverse model:

$$p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t) = \mathcal{N}\big(\boldsymbol{x}_{t-1};\ \boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t),\ \sigma_t^2 \mathbf{I}\big)$$

Training objective: for each t, minimise

$$\mathcal{D}_{\mathrm{KL}}\big(q(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \boldsymbol{x}_0)\,\big\|\,p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t)\big)$$

Both distributions are Gaussian, so the KL has a closed form.

DDPM: ϵ-Prediction and Training Loss

Parameterise the mean via a noise-prediction network $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)$:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t)\right)$$

Intuition: the network is a *noise detective*. Given $\boldsymbol{x}_t$, it estimates $\boldsymbol{\epsilon}$; subtracting it moves closer to $\boldsymbol{x}_0$.

Since $\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$, predicting $\boldsymbol{\epsilon}$ is equivalent to predicting $\boldsymbol{x}_0$.

DDPM Simplified Training Loss

$$\mathcal{L}_{\text{simple}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\, \boldsymbol{x}_0,\, \boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,\ t)\|^2\right]$$

where $\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $t \sim \mathcal{U}\{1, \ldots, T\}$.

Relationship to ELBO: The full ELBO decomposes into a prior term, a reconstruction term, and $T-1$ diffusion terms. The prior and reconstruction terms are negligible; $\mathcal{L}_{\text{simple}}$ is a reweighted version of the diffusion terms that works better in practice.

DDPM: Sampling Algorithm

Reverse (iterative denoising):

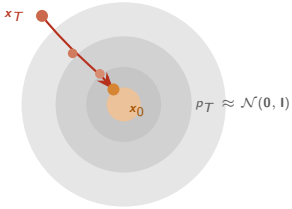
Sample $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. For $t = T, T-1, \ldots, 1$:

$$\boldsymbol{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\boldsymbol{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\right) + \sigma_t \boldsymbol{z}$$

where $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $t > 1$, $\boldsymbol{z} = \mathbf{0}$ for $t = 1$.

Each step:

1. Estimate noise $\hat{\epsilon} = \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$
2. Recover clean estimate $\hat{\boldsymbol{x}}_0$
3. Re-noise to level $t-1$



progressive denoising
coarse $\longrightarrow$ fine

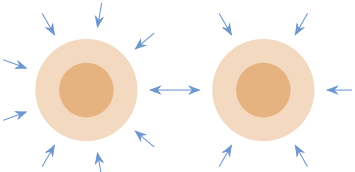
Note: DDPM needs $T \approx 1000$ steps (small β_t required for Gaussian approximation to hold). DDIM and other solvers reduce this substantially.

Score Function and Energy-Based Models

Energy-Based Model (EBM):

$$p_\phi(\boldsymbol{x}) = \frac{\exp(-E_\phi(\boldsymbol{x}))}{Z_\phi}, \quad Z_\phi = \int \exp(-E_\phi)\, d\boldsymbol{x}$$

Problem: Z_ϕ is *intractable* — MLE requires sampling from p_ϕ.



$\boldsymbol{s}(\boldsymbol{x}) = \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x})$
points toward high-density regions

The Score Function

For a density $p(\boldsymbol{x})$, the *score* is

$$\boldsymbol{s}(\boldsymbol{x}) := \nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}), \quad \boldsymbol{s} : \mathbb{R}^D \to \mathbb{R}^D$$

Key: the partition function cancels!

$$\nabla_{\boldsymbol{x}} \log p_\phi(\boldsymbol{x}) = -\nabla_{\boldsymbol{x}} E_\phi(\boldsymbol{x})$$

No need to compute Z_ϕ.

Langevin dynamics:

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n + \eta\, \boldsymbol{s}(\boldsymbol{x}_n) + \sqrt{2\eta}\, \boldsymbol{\epsilon}_n$$

Converges to p as $\eta \to 0$, steps $\to \infty$.

Score Matching: Learning the Score

Idea: train $\boldsymbol{s}_\phi(\boldsymbol{x})$ to approximate $\nabla_{\boldsymbol{x}} \log p_{\text{data}}(\boldsymbol{x})$ directly, bypassing the intractable Z_ϕ.

Naive objective (intractable — true score unknown):

$$\mathcal{L}_{SM} = \tfrac{1}{2}\mathbb{E}_{p_{\text{data}}}\left[\|\boldsymbol{s}_\phi(\boldsymbol{x}) - \nabla_{\boldsymbol{x}} \log p_{\text{data}}(\boldsymbol{x})\|^2\right]$$

Hyvärinen's Tractable Form (2005)

Integration by parts gives an equivalent objective:

$$\tilde{\mathcal{L}}_{SM}(\phi) = \mathbb{E}_{p_{\text{data}}}\left[\mathrm{Tr}(\nabla_{\boldsymbol{x}} \boldsymbol{s}_\phi) + \tfrac{1}{2}\|\boldsymbol{s}_\phi(\boldsymbol{x})\|^2\right] + C$$

Same minimiser — **no true score needed**.
Cost: $\mathcal{O}(D^2)$ Jacobian trace (expensive for large D).

Denoising Score Matching (Vincent 2011)

Perturb data: $\tilde{\boldsymbol{x}} = \boldsymbol{x} + \sigma\boldsymbol{\epsilon}, \;\; \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Conditional score has closed form:

$$\nabla_{\tilde{\boldsymbol{x}}} \log p_\sigma(\tilde{\boldsymbol{x}}|\boldsymbol{x}) = \frac{\boldsymbol{x} - \tilde{\boldsymbol{x}}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

DSM loss (no Jacobian needed):

$$\mathcal{L}_{DSM}(\phi; \sigma) = \tfrac{1}{2}\mathbb{E}_{\boldsymbol{x},\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{s}_\phi(\tilde{\boldsymbol{x}}; \sigma) + \frac{\boldsymbol{\epsilon}}{\sigma}\right\|^2\right]$$

Theorem: $\mathcal{L}_{SM} = \mathcal{L}_{DSM} + C$. Same minimiser: $\boldsymbol{s}^*(\,\cdot\,; \sigma) = \nabla_{\boldsymbol{x}} \log p_\sigma(\boldsymbol{x})$.

DDPM $\equiv$ Score Matching: The Deep Connection

Score and noise prediction are equivalent!

For the DDPM forward process, Tweedie's formula gives:

$$\mathbb{E}[\boldsymbol{x}_0|\boldsymbol{x}_t] = \frac{1}{\sqrt{\bar{\alpha}_t}}(\boldsymbol{x}_t + (1-\bar{\alpha}_t)\,\boldsymbol{s}(\boldsymbol{x}_t, t))$$

Since $\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$, the score and the noise predictor are related by:

$$\boxed{\boldsymbol{s}^*(\boldsymbol{x}_t, t) = -\frac{\boldsymbol{\epsilon}^*(\boldsymbol{x}_t, t)}{\sqrt{1-\bar{\alpha}_t}}}$$

Three Equivalent Predictions

Target	Network	Use case
Score $\boldsymbol{s}^*$	$\boldsymbol{s}_\theta$	Langevin, SDE
Noise $\boldsymbol{\epsilon}^*$	$\boldsymbol{\epsilon}_\theta$	DDPM loss
Clean $\boldsymbol{x}_0^*$	$\boldsymbol{x}_\theta$	x-prediction

All three are deterministic reparameterisations of each other via the forward process relation $\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$.

Training any one of them is sufficient; the other two follow automatically.

Flow Matching: The ODE Perspective

Goal: find a time-varying velocity field $\boldsymbol{v}_\phi(\boldsymbol{x}, t)$ such that integrating the ODE

$$\frac{d\boldsymbol{x}(t)}{dt} = \boldsymbol{v}_\phi(\boldsymbol{x}(t), t), \quad t \in [0, 1]$$

transports samples from a simple prior $p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I})$ to the data distribution $p_1 = p_{\text{data}}$.

Advantages over diffusion:

- Continuous, smooth trajectories
- Fewer ODE steps at inference
- Direct connection to Score SDE via PF-ODE

Continuity Equation

The density p_t induced by the flow satisfies:

$$\frac{\partial p_t}{\partial t} + \nabla_{\boldsymbol{x}} \cdot (p_t \, \boldsymbol{v}_t) = 0$$

Any $\boldsymbol{v}_t$ satisfying this equation correctly transports mass from p_0 to p_1.

Challenge: the *marginal* velocity $\boldsymbol{v}_t(\boldsymbol{x}) = \mathbb{E}[\dot{\boldsymbol{x}} \mid \boldsymbol{x}_t = \boldsymbol{x}]$ satisfying the continuity equation is intractable in general.

Key idea: define tractable *conditional* paths and regress onto their velocities — no ODE simulation required during training.

Conditional Flow Matching

FM loss (intractable marginal velocity $\boldsymbol{v}_t$):

$$\mathcal{L}_{FM}(\phi) = \mathbb{E}_{t\sim\mathcal{U}[0,1],\ \boldsymbol{x}\sim p_t}\left[\|\boldsymbol{v}_\phi(\boldsymbol{x},t) - \boldsymbol{v}_t(\boldsymbol{x})\|^2\right]$$

CFM: The Key Insight

Introduce $\boldsymbol{z} = \boldsymbol{x}_1 \sim p_{\text{data}}$ and define a *conditional* path $p_t(\boldsymbol{x}|\boldsymbol{z})$ and velocity $\boldsymbol{v}_t(\boldsymbol{x}|\boldsymbol{z})$ with **closed form**.

CFM loss:

$$\mathcal{L}_{CFM}(\phi) = \mathbb{E}_{t,\,\boldsymbol{z},\ \boldsymbol{x}\sim p_t(\cdot|\boldsymbol{z})}\left[\|\boldsymbol{v}_\phi(\boldsymbol{x},t) - \boldsymbol{v}_t(\boldsymbol{x}|\boldsymbol{z})\|^2\right]$$

Theorem: $\mathcal{L}_{FM}(\phi) = \mathcal{L}_{CFM}(\phi) + C$.
Same minimiser: $\boldsymbol{v}^*_\phi(\boldsymbol{x},t) = \boldsymbol{v}_t(\boldsymbol{x})$.

Conditioning trick — a unifying principle:

- DDPM: condition on $\boldsymbol{x}_0$ to get tractable KL targets
- DSM: condition on clean data point to get closed-form score
- CFM: condition on $\boldsymbol{x}_1$ to get closed-form velocity

Gaussian (Linear) Path

Set $p_t(\boldsymbol{x}|\boldsymbol{z}) = \mathcal{N}(\boldsymbol{x};\ t\,\boldsymbol{z},\ \sigma^2\mathbf{I})$ with $\boldsymbol{z} = \boldsymbol{x}_1 \sim p_{\text{data}}$:

$$\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1, \quad \boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0},\mathbf{I})$$

$$\boldsymbol{v}_t(\boldsymbol{x}_t|\boldsymbol{x}_0,\boldsymbol{x}_1) = \boldsymbol{x}_1 - \boldsymbol{x}_0$$

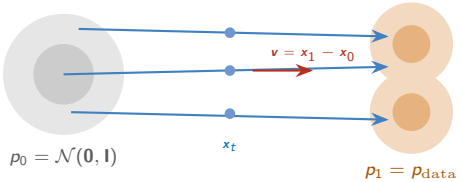
Gaussian Flow Matching: Linear Interpolation

Linear (affine) path:

$$\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1, \quad \boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),\ \boldsymbol{x}_1 \sim p_{\text{data}}$$

Conditional velocity (derivative of path):

$$\boldsymbol{v}_t(\boldsymbol{x}_t|\boldsymbol{x}_0, \boldsymbol{x}_1) = \frac{d\boldsymbol{x}_t}{dt} = \boldsymbol{x}_1 - \boldsymbol{x}_0 \quad \text{(constant!)}$$



$x_t = (1-t)x_0 + tx_1$
straight-line paths

Gaussian FM Training Loss

$$\mathcal{L}_{CFM}(\phi) = \mathbb{E}_{t,\,\boldsymbol{x}_0,\,\boldsymbol{x}_1}\left[\|\boldsymbol{v}_\phi(\boldsymbol{x}_t, t) - (\boldsymbol{x}_1 - \boldsymbol{x}_0)\|^2\right]$$

Remarkably simple: just predict the direction $\boldsymbol{x}_1 - \boldsymbol{x}_0$.

Score SDE: The Continuous-Time View

Continuous-time limit of the DDPM forward process is an SDE (Song et al. 2021):

$$d\boldsymbol{x} = f(t)\boldsymbol{x}\,dt + g(t)\,d\boldsymbol{w}$$

$\boldsymbol{w}$ = Brownian motion; $f(t)$ controls drift, $g(t)$ controls noise injection.

Reverse-time SDE (Anderson, 1982):

$$d\boldsymbol{x} = \left[f(t)\boldsymbol{x} - g^2(t)\,\nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x})\right] dt + g(t)\,d\bar{\boldsymbol{w}}$$

The score $\nabla_{\boldsymbol{x}} \log p_t$ drives the reverse process.

VP-SDE (corresponds to DDPM):

$$f(t) = -\tfrac{1}{2}\beta(t), \qquad g(t) = \sqrt{\beta(t)}$$

Probability Flow ODE (PF-ODE)

A *deterministic* process with the same marginals $\{p_t\}$:

$$\frac{d\boldsymbol{x}}{dt} = f(t)\boldsymbol{x} - \tfrac{1}{2}g^2(t)\,\nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x})$$

The score determines the ODE velocity — directly connecting Score SDE to Flow Matching.

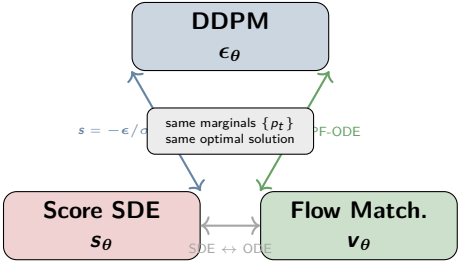
Fokker–Planck equation:

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot (f(t)\boldsymbol{x}\,p_t) + \frac{g^2(t)}{2}\Delta p_t$$

One Model, Three Languages

Core Equivalence (Gaussian FM, $\alpha_t = 1 - t$)

Approach	Predicts	Relation to score
DDPM	$\boldsymbol{\epsilon}^*$	$\boldsymbol{s}^* = -\boldsymbol{\epsilon}^*/\sigma_t$
Score SDE	$\boldsymbol{s}^* = \nabla \log p_t$	(definition)
PF-ODE / FM	$\boldsymbol{v}^*$	$\boldsymbol{v}^* = f\boldsymbol{x} - \frac{g^2}{2}\boldsymbol{s}^*$



All three are deterministic reparameterisations of the *same optimal predictor*.

Shared principle (appears in all three):

1. Define a tractable *conditional* target
2. Show: conditional loss = marginal loss + constant
3. Train a single network on the conditional regression

Why Do These Models Succeed?

Structural advantages:

1. **Progressive generation:** the noise schedule creates a curriculum from "easy" (coarse) to "hard" (fine) tasks
2. **Well-conditioned objectives:** each step reduces to a regression problem with stable, known targets
3. **No adversarial dynamics:** unlike GANs, training is stable; no mode collapse or oscillations
4. **Flexible architectures:** any backbone (UNet, ViT/DiT) can serve as the denoiser or velocity predictor

Key milestones:

Model	Year	Contribution
DDPM	2020	Simplified noise-regression loss
DDIM	2021	Deterministic, fast sampling
Score SDE	2021	Continuous SDE framework
LDM	2022	Latent-space diffusion
FM / RF	2022	Straight-line ODE paths
DiT	2023	Transformer backbone

Applications: text-to-image (Stable Diffusion, DALL-E 3), video (Sora), audio, molecular design (AlphaFold 3, RFDiffusion).